QP CODE: 25901160

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU – UGP (HONOURS) REGULAR EXAMINATION MARCH 2026

FOURTH SEMESTER

Discipline Specific Core Course (DSC) –MG4DSCCND200

CLINICAL NUTRITION II

(2024 ADMISSION ONWARDS)

Duration: 1Hour 30 minutes Maximum Marks: 50

Remember(K),Understand(U),Apply(A),Analyse(An),Evaluate(E),Create(C),Interest (I),Appreciation(Ap),and Skill(S)

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries 1 mark

1. c) Alcohol abuse
2. b) Ascites
3. a) LDL
4. b) 500 ml + urine output
5. b) Fluid accumulation
6. a) 2-6g

Part B

Short Answer Questions

Answer any **2** questions.

Each question carries **2** marks.

7. **Discuss the formation of indinavir stones.**

- They are soft, gelatinous and radioluscent stones.
- Treatment of HIV infection with protease inhibitors led to formation of indinavir stones.
- The stones have decreased solubility with low urine volume and low pH.
- Hypocitraturia is seen in all indinavir stone patients. ***(½ mark for each point)***

8. Discuss about ischemic heart disease.

- Condition when heart muscle is weakened as a result of heart attack or CAD.
- Arteries that supply blood to heart becomes narrowed causing damage to heart muscle
- The left ventricle becomes enlarged, weakened and dilated.
- Inhibits heart's ability to pump blood properly which leads to heart failure.

(½ mark for each point)

9. Discuss medicine induced diarrhoea

- These are frequent in hospitalized and long-term care patients.
- Medications such as lactulose and sodium polystyrene sulfonate with sorbitol create increased bowel movements.
- Broad-spectrum antibiotics decrease the number of bacteria in the bowel and may result in increased osmotically active molecules, reduced absorption of electrolytes and water, and diarrhoea.
- The treatment of Escherichia coli and several other organisms has been implicated in antibiotic-associated diarrhoea (AAD).

(½ mark for each point)

10. Define Lymphocytic oesophagitis

- Lymphocytic esophagitis is an uncommon oesophageal condition in which there are an increased number of lymphocytes in the lining of the oesophagus.

(2 marks for definition)

Part C

Short Essay

Answer any **4 questions**

Each question carries **5** marks.

11. Review dyspepsia, causes and symptoms.

The term Dyspepsia is used for a condition in which there is difficulty in digestion of food. ***(I mark for definition)***

Causes of Dyspepsia

- It can be caused by a number of conditions, like ulcers, gall bladder diseases, pancreatitis, hepatitis and irritable bowel syndrome. The disease can occur due to organic diseases of gastrointestinal tract, cardiovascular or kidney disease or malignant disease.
- In the absence of organic reasons, rapid eating, inadequate chewing, swallowing air through food, ingestion of undercooked foods or fried foods, gas forming vegetables, protein-rich animal foods, absence of bulk in the diet produce the presence of air in the stomach or intestine.

- The most common cause of dyspepsia is stomach acid. Acid can irritate the lining of the stomach, oesophagus or small intestine, causing that familiar burning or painful sensation.
- The major causes of dyspepsia are irregular food habits, heavy, late-night dinner, very spicy foods, beverages and alcohol, excessive physical or mental strain, excessive use of medicines, too much fried foods, sugars, sweets and starches, drinking excessive fluids in between the meals causes bloating of the stomach.

(2 marks - ½ mark for each point)

Symptoms of Dyspepsia

Symptoms of this disorder are bitter taste in the mouth, heart burn, nausea, loss of appetite, distention, belching, anorexia, epigastric distress or pain, discomfort or pain in the stomach, vomiting and cardio spasm. ***(2 marks if any eight listed)***

12. Discuss the role of dietary deficiency in liver damage

- Fatty changes in liver in kwashiorkor may be due to low protein intake and reduced capacity to produce lipoproteins.
- The liver is often grossly enlarged in children with kwashiorkor, containing 30-50 per cent fat.
- Intrahepatic accumulation of fat results due to excess delivery of fatty acids to the liver or enhanced lipogenesis.
- This is combined with impaired lipid transport from the liver secondary to apoprotein deficiency. When protein is supplied fat clears from the liver.
- Fatty changes are common whenever there is high proportion of fat - in uncontrolled diabetes, starvation, obesity and when too much CHO are infused through IV feed.

(1mark for each point)

13. Explain the category I risk factors of CVD.

- **Cigarette smoking** – enhance endothelial damage, increase heart rate and BP, increase LDL, decrease HDL, constrict blood vessel and promote thrombus formation.
- **High saturated fat, high cholesterol diet –** increase cholesterol, factor VII, and atheroma formation. Saturated fat reduce LDL and VLDL receptor formation and inhibit antiaggregatory prostacyclin.
- **Elevated LDL –** develops atherosclerosis
- **Hypertension –** leads to enhanced endothelial cell damage and target organ involvement including retinopathy and renal insufficiency. Hypertension is associated with obesity, blood lipid abnormalities, insulin resistance and glucose intolerance.

- **Thrombogenic factors -** leads to development of intracoronary thrombosis. High fibrinogen concentrations, plasminogen activator-inhibitor I, platelet aggregability and factor VII concentration increases leading to CVD. ***(All 5 points – 1 mark each)***

14. **Explain the causes and symptoms of acute renal failure.**

Causes

- Loss of blood – as blood flow to kidney decreases in case of accidents, ulcers, haemorrhages and at the time of delivery
- Loss of plasma in burns and crush injuries
- Loss of fluid in diarrhoea, vomiting, diabetic coma
- General anaesthesia and surgical operations reduce renal blood flow
- Acute haemolytic disorders and Serious infections produce shock and reduce renal blood flow ***(2.5 marks – ½ mark each for each point)***

Symptoms

- Low urine volume as 20 – 200 ml/day
- Increased serum urea nitrogen and creatinine due to accumulation of waste products of protein metabolism
- Increased serum potassium due to release from increased protein breakdown to provide calories
- Increased phosphate and sulphate with decreased sodium, calcium and base bicarbonate
- Lethargy, nausea, vomiting diarrhoea
- Death due to potassium and water intoxication

(2.5 marks – ½ mark each for each point)

15. Explain hypercholesterolemia and its clinical features.

- Increased blood cholesterol level above 210 mg/dl.

Clinical features

- Increased risk of CHD with a heart attack due to myocardial infarction.
- Xanthomata – swellings or small tumours containing cholesterol and some triglycerides
- Common sites of xanthomata are tendons, especially Achilles tendon and tendons on back of hand and skin around eye known as xanthelasma
- Corneal arcus is seen at early age due to hypercholesterolemia.

(All 5 points – 1 mark each)

16. Differentiate diverticulosis and diverticulitis.

Diverticulosis	**Diverticulitis**
Diverticulosis is the structural presence of	Diverticulitis is the inflammation or infection

pouches (diverticula) in the colon wall, often found in people over 50.	of these pouches.
Diverticulosis is caused by pressure on weak spots in the colon.	Diverticulitis occurs when these pouches become inflamed, sometimes due to bacteria.
Diverticulosis is typically asymptomatic.	Diverticulitis causes acute abdominal pain (often lower-left side), fever, and nausea, vomiting, and, rarely, bleeding.
Diverticulosis is managed by diet (high-fiber) and, in some cases, lifestyle changes.	Diverticulitis requires medical intervention, such as antibiotics, a liquid diet, or, in severe cases, hospitalization and surgery.

(2marks for the first point, 1 marks each for the rest of the points)

Part D

Essay

Answer any **2** questions

Each question carries **10** marks

17. Explain haemodialysis and its dietary management

- Haemodialyser is used to remove waste from body and maintain the chemical balance
- It consists of two compartments separated by a thin porous membrane
- One compartment has patients blood and the other has the cleaning solution called dialysate.
- Plastic tubing attached to needles connect the patient to the machine
- Smaller waste products like urea, creatinine and excess water is washed off.
- The blood cells, protein and other required substances remain and enters the patients body
- Needed substances like calcium and dextrose can be added to dialysate.
- Haemodialysis is performed with the help of an artificial fistula or graft.

(4 marks – 1/4 mark for each point)

Dietary management

- Energy – 35 kcal/kg body weight
- Protein – 1.1 g/kg body weight
- Dietary salt : 3-4 g/day
- Potassium : 50-7- mEq/day
- Phosphorus – 1 g/day
- Water – according to urine output of patient ***(6 marks – 1 mark for each point)***

18. Explain the types, symptoms and dietary management in hypertension

Types

- Primary hypertension – when genetic predisposition causes hypertension
- Secondary hypertension – hypertension caused by medication or illness
- Essential hypertension – cause of hypertension is unknown

- Mild hypertension – diastolic pressure is 90 – 104 mm Hg
- Moderate hypertension - diastolic pressure is 105 – 119 mm Hg
- Severe hypertension – diastolic pressure is 120 – 130 mm Hg

(3 marks – ½ marks each for each point)

Symptoms

- Headache, dizziness, impaired vision, failing memory, shortness of breath, pain over heart, gastrointestinal disturbance and unexplained tiredness, bulges and ruptures of blood vessel wall of heart, eyes or kidney ***(2 marks – ¼ marl for each point)***

Dietary management

- Energy – 20 kcal/kg of ideal body weight for sedentary and 25 kcal/kg of body weight for moderate worker
- Protein – 60 g/day is advised. In severe conditions 20 g as a temporary measure is advised as per Kempners diet
- Fat – 20 g vegetable oil. Omega 3 helps in regulating blood pressure.
- Sodium – restricted to 2-3 g/day. Increased sodium may lead to increased hypertension
- Potassium – 3500 mg/day. Low level of potassium may retain sodium and water leading to elevated BP (***5 marks – 1 mark for each point).***

19. Summarize the types, causes, symptoms and dietary management of cholecystitis.

Cholecystitis is the inflammation of gall bladder. It is usually caused by gallstones obstructing the bile ducts preventing the passage of bile into duodenum causing a backflow of bile.

(1 mark)

Types

- ***Acute Cholecystitis*** **-** This can occur without stones mainly in critically ill patients or when the gall bladder and its bile are stagnant.
- ***Chronic Cholecystitis*** **-** Chronic cholecystitis is long-standing inflammation of the gallbladder. It is caused by repeated, mild attacks of acute cholecystitis.

(2 Marks – 1 mark for each type)

Causes/Risk factors

- **Dietary factors** – high fat diet over a long period of time, cholesterol rich foods, high sugar and intake of refined carbohydrate, low fibre diet, rapid losing of weight.
- **Metabolic factors** - obesity, high waist hip ratio, insulin resistance, diabetes and hyperlipidemia.
- **Physiological factors** - female gender, family history of gallstones on the mother's side of the family, older age, pregnancy.
- **Medical conditions** - Crohn’s disease. ***(2 marks - 1/2 marks for each point)***

Symptoms

Severe pain in upper right or centre of abdomen, pain that spreads to right shoulder or back, tenderness over abdomen when it is touched, abdominal bloating, tenderness on the

upper-right hand side of the abdomen nausea, vomiting, fever, little or no appetite, sweating, blood test may reveal a high white blood cell count.

(1 mark)

Dietary management

Principles of diet - Adequate calorie, protein, carbohydrate, low fat, moderate to high fibre, adequate fluid. ***(1 mark)***

Nutritional requirements

- **Calorie** –Consume only the required calorie to maintain a healthy weight.
- **Protein** –The recommended intake is 60 gm. Choose lean protein sources.
- **Fat** – Initially restrict fat to 20-30gm/day. When the patient can tolerate, gradually increase to 50-60g/day. Fat should be less than 25-30% of total calories.
- **Fibre** – Include moderate fibre, which helps in digestion and reduce cholesterol absorption.
- **Vitamins** – Supplement fat soluble vitamins.
- **Fluid** – Adequate fluid is essential to keep the bile thin. It helps prevent stagnation.

(3 Marks- ½ marks for each point)

20. Explain the symptoms and dietary management of rheumatic heart disease

Symptoms

- Chest pain, excessive fatigue, heart palpitation, thumping sensation in the chest, shortness of breath, swollen ankles, wrists or stomach. (**2 marks)**

Dietary management

- Energy – 1000 – 1200 kcal / day. In case of obese patients loss of weight helps in better functioning of heart
- Fat- 20 per cent of total calories. PUFA to saturated fat ratio 0.8:1. PUFA promotes esterification of cholesterol to easily utilisable forms like thromboxanes and prostacyclins which prevent thrombus formation
- Carbohydrates – include complex carbohydrates. High CHO can increase triglyceride levels. Restricting to 60 per cent of energy intake can reduce risk of CVD
- Protein – normal allowances are suggested. Animal protein is not suggested.
- Vitamin B6, B12, folic acid supplementation reduce CVD risk related to homocysteine risk.
- Vitamin C is involved in regulating cholesterol metabolism and maintaining structure of blood vessels. Vitamin C activates adrenals and reduce cholesterol
- Sodium – restricted to 1600 -2300 mg.

- Potassium and magnesium helps to relax muscles, keep a healthy tone and maintain steady heart rate.

(8 marks – 1 marks/point)